A guide to the arithmetic certificate for $K = \mathbf{Q}(\sqrt{-145367147})$, $p = 5$

Generated by `tools/certificate_guide.gp`

What this document is

This guide restates the content of `certificate.gp` in the notation of Section 2.5 of the paper. Each of the 18 entries certifies one value $D_x(e_j)$ of a secondary norm operator: it stores the unramified cyclic quintic L_x/K, the normalized generator σ_x, the pair (a', J) representing the torsion class e_j, the auxiliary divisor I', the compactly represented element t, a residue prime for the sign check, and the expected norm class $[N_x(I')]$. The two defining equations are

$$(1 - \sigma_x)^2 I' + \operatorname{div}(t) + i_x(J) = 0, \qquad\qquad N_x(t) = a'.$$

Facts marked **Checked** are recomputed exactly by the generating script; the full verification is performed by the shared verifier `certificate/verify_certificate.c` as described in the README files of the `certificate` directory.

Notation and conventions

Ideals as matrices. An ideal of the ring of integers is stored by its *Hermite normal form* (HNF): an upper-triangular integer matrix whose columns are the coordinates, with respect to the fixed integral basis, of a $\mathbf{Z}$-basis of the ideal. For example, the ideal $J = \left(\begin{smallmatrix} 243 & 146 \\ 0 & 1 \end{smallmatrix}\right)$ of Entry 1 below, over K with integral basis $(1, s)$, denotes the ideal $\mathbf{Z}\,243 + \mathbf{Z}\,(146 + s)$. The determinant of the matrix is the norm of the ideal. Elements are likewise given by their coordinate vectors in the integral basis.

Which integral basis. The one PARI returns for the field in question – and that basis is LLL-reduced, so it is not canonical: another machine may reduce differently, and the same coordinate vector would then denote a different algebraic number. The certificate therefore records the basis it was written against, as its last field per entry and in the header for K. The verifier expresses the stored basis in its own, checks that the resulting matrix is integral with determinant ± 1 – so that both bases span the same ring of integers – and converts the coordinates before checking anything. Where the two bases agree, that matrix is the identity and nothing changes.

Characters. The header fixes three generators $\kappa_1, \kappa_2, \kappa_3$ of $\mathrm{Cl}(K)$ (as HNF ideals). A character x on $\mathrm{Cl}(K)/5$ is recorded by its value vector $(x(\bar\kappa_1), x(\bar\kappa_2), x(\bar\kappa_3))$; thus $(1, 0, 0)$ is the character dual to $\bar\kappa_1$. The column number j records which basis element e_j of $\mathrm{Cl}(K)[5]$ the stored pair (a', J) represents.

Automorphisms. σ_x is a field automorphism of L_x. Writing θ for the chosen root of the stored absolute polynomial, so that $L_x = \mathbf{Q}(\theta)$, the automorphism is determined by the single value $\sigma_x(\theta)$, and the certificate stores the coordinate vector of $\sigma_x(\theta)$ in the integral basis of L_x.

Base field data

The base field is $K = \mathbf{Q}(s)$ with $s^2 - s + 36341787 = 0$, of discriminant -145367147. Class group invariants $(125 \quad 5 \quad 5)$, class number 3125; the three stored generator ideals (HNF) are $\begin{pmatrix} 11 & 1 \\ 0 & 1 \end{pmatrix}$, $\begin{pmatrix} 3891 & 2759 \\ 0 & 1 \end{pmatrix}$, $\begin{pmatrix} 5949 & 3737 \\ 0 & 1 \end{pmatrix}$, and the torsion units are generated by -1 of order 2.

Entry 1: character a, column e_1

Prescribed character vector $(1 \quad 0 \quad 0)$, i.e. the values of x on the classes of the three stored generators of $\mathrm{Cl}(K)$.

The extension and its normalized generator

Relative polynomial of L_x over K:

$$y^5 - 2y^4 + (-s - 1461)\, y^3 + (40s - 22841)\, y^2 + (-496s + 1318012)\, y + (2091s - 10470351)$$

Absolute polynomial of L_x over $\mathbf{Q}$ (degree 10):

```
y^10 - 4*y^9 - 2919*y^8 - 39796*y^7 + 41204581*y^6 - 2866846824*y^5
+ 90908762347*y^4 - 1623567054940*y^3 + 17234257097955*y^2 - 1029748
98471300*y + 268502861365407
```

The automorphism σ_x is determined by the image of the primitive element θ: the coordinates of $\sigma_x(\theta)$ in the integral basis of L_x are $(0 \quad 0 \quad 0 \quad -1 \quad 0 \quad 0 \quad 0 \quad 1 \quad 0 \quad 0)$.

The pair (a', J) representing the torsion class

$$a' = -\frac{302351}{282429536481} + (\frac{26}{847288609443})\, s, \qquad J = \begin{pmatrix} 243 & 146 \\ 0 & 1 \end{pmatrix}.$$

$N(J) = 243$. **Checked:** $(a')\, J^5 = \mathcal{O}_K$, equivalently $\mathrm{div}(a') + 5J = 0$: pass.

The auxiliary pair (t, I')

I' is the ideal of L_x with the following HNF matrix in the integral basis of the absolute model; its absolute norm is $148071703930416 = 2^4 \cdot 3^4 \cdot 7^2 \cdot 2331690979$:

$$\begin{pmatrix}
1762758380124 & 295958620524 & 698920297134 & 1022332005717 & 1706469471437 & 857437419920 & 173977642215 & 339849787429 & 494138011360 & 983301729799 \\
0 & 28 & 16 & 17 & 13 & 17 & 25 & 8 & 14 & 10 \\
0 & 0 & 3 & 2 & 2 & 1 & 2 & 2 & 0 & 0 \\
0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1
\end{pmatrix}$$

The element t is stored in compact form as a product of 226 factors with signed exponents of absolute value up to 1842929065801238; it is never expanded (its principal ideal and residues are evaluated factor by factor).

Sign check and certified value

Residue prime: the stored prime ideal above $\ell = 11$ with residue degree 1; there the quotient $N_x(t)/a'$ reduces to the residue of $+1$. Certified value: $D_x(e_1) = (0 \quad 2 \quad 1)$ in the coordinates of $\mathrm{Cl}(K)/5$.

Entry 2: character a, column e_2

Prescribed character vector $(1 \quad 0 \quad 0)$, i.e. the values of x on the classes of the three stored generators of $\mathrm{Cl}(K)$.

The extension and its normalized generator

Relative polynomial of L_x over K:

$$y^5 - 2y^4 + (-s - 1461)\, y^3 + (40s - 22841)\, y^2 + (-496s + 1318012)\, y + (2091s - 10470351)$$

Absolute polynomial of L_x over $\mathbf{Q}$ (degree 10):

```
y^10 - 4*y^9 - 2919*y^8 - 39796*y^7 + 41204581*y^6 - 2866846824*y^5
+ 90908762347*y^4 - 1623567054940*y^3 + 17234257097955*y^2 - 1029748
98471300*y + 268502861365407
```

The automorphism σ_x is determined by the image of the primitive element θ: the coordinates of $\sigma_x(\theta)$ in the integral basis of L_x are $(0 \quad 0 \quad 0 \quad -1 \quad 0 \quad 0 \quad 0 \quad 1 \quad 0 \quad 0)$.

The pair (a', J) representing the torsion class

$$a' = -\frac{107118472}{297293147712326817} + \left(\frac{147313}{891879443136980451}\right) s, \qquad J = \begin{pmatrix} 3891 & 2759 \\ 0 & 1 \end{pmatrix}.$$

$N(J) = 3891$. **Checked:** $(a')\, J^5 = \mathcal{O}_K$, equivalently $\mathrm{div}(a') + 5J = 0$: pass.

The auxiliary pair (t, I')

I' is the ideal of L_x with the following HNF matrix in the integral basis of the absolute model; its absolute norm is $323079881298417 = 3^7 \cdot 37 \cdot 3992633143$:

$$\begin{pmatrix}
1329546836619 & 488313146556 & 763379049792 & 995756518875 & 275476837032 & 764482833148 & 482091980565 & 150952630306 & 418172374833 & 324639002923 \\
0 & 9 & 0 & 1 & 0 & 6 & 8 & 8 & 0 & 8 \\
0 & 0 & 3 & 2 & 0 & 1 & 2 & 0 & 1 & 2 \\
0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 3 & 2 & 0 & 1 & 0 & 2 \\
0 & 0 & 0 & 0 & 0 & 3 & 0 & 2 & 1 & 1 \\
0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1
\end{pmatrix}$$

The element t is stored in compact form as a product of 211 factors with signed exponents of absolute value up to 2568514874133254; it is never expanded (its principal ideal and residues are evaluated factor by factor).

Sign check and certified value

Residue prime: the stored prime ideal above $\ell = 11$ with residue degree 1; there the quotient $N_x(t)/a'$ reduces to the residue of $+1$. Certified value: $D_x(e_2) = (0 \quad 2 \quad 3)$ in the coordinates of $\mathrm{Cl}(K)/5$.

Entry 3: character a, column e_3

Prescribed character vector $(1 \quad 0 \quad 0)$, i.e. the values of x on the classes of the three stored generators of $\mathrm{Cl}(K)$.

The extension and its normalized generator

Relative polynomial of L_x over K:

$$y^5 - 2y^4 + (-s - 1461)\, y^3 + (40s - 22841)\, y^2 + (-496s + 1318012)\, y + (2091s - 10470351)$$

Absolute polynomial of L_x over $\mathbf{Q}$ (degree 10):

```
y^10 - 4*y^9 - 2919*y^8 - 39796*y^7 + 41204581*y^6 - 2866846824*y^5
+ 90908762347*y^4 - 1623567054940*y^3 + 17234257097955*y^2 - 1029748
98471300*y + 268502861365407
```

The automorphism σ_x is determined by the image of the primitive element θ: the coordinates of $\sigma_x(\theta)$ in the integral basis of L_x are $(0 \quad 0 \quad 0 \quad -1 \quad 0 \quad 0 \quad 0 \quad 1 \quad 0 \quad 0)$.

The pair (a', J) representing the torsion class

$$a' = -\frac{567178771}{2483696869417001583} + (-\frac{354041}{7451090608251004749})\, s, \qquad J = \begin{pmatrix} 5949 & 3737 \\ 0 & 1 \end{pmatrix}.$$

$N(J) = 5949$. **Checked:** $(a')\, J^5 = \mathcal{O}_K$, equivalently $\operatorname{div}(a') + 5J = 0$: pass.

The auxiliary pair (t, I')

I' is the ideal of L_x with the following HNF matrix in the integral basis of the absolute model; its absolute norm is $49278904535232 = 2^6 \cdot 3^7 \cdot 37 \cdot 73 \cdot 130349$:

$$\begin{pmatrix} 114071538276 & 14127364152 & 54932617404 & 50709401370 & 32269400549 & 355607676 & 84245706094 & 40054680368 & 12560127955 & 2119595412 \\ 0 & 12 & 0 & 6 & 9 & 3 & 8 & 7 & 3 & 4 \\ 0 & 0 & 6 & 4 & 2 & 3 & 3 & 3 & 5 & 3 \\ 0 & 0 & 0 & 2 & 0 & 0 & 1 & 1 & 1 & 1 \\ 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 3 & 2 & 2 & 0 & 1 \\ 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 \end{pmatrix}$$

The element t is stored in compact form as a product of 234 factors with signed exponents of absolute value up to 2131474240736027; it is never expanded (its principal ideal and residues are evaluated factor by factor).

Sign check and certified value

Residue prime: the stored prime ideal above $\ell = 11$ with residue degree 1; there the quotient $N_x(t)/a'$ reduces to the residue of $+1$. Certified value: $D_x(e_3) = (0 \quad 0 \quad 0)$ in the coordinates of $\mathrm{Cl}(K)/5$.

Entry 4: character b, column e_1

Prescribed character vector $(0 \quad 1 \quad 0)$, i.e. the values of x on the classes of the three stored generators of $\mathrm{Cl}(K)$.

The extension and its normalized generator

Relative polynomial of L_x over K:

$$y^5 + (s - 3032)\, y^3 + (36s - 80152)\, y^2 + (341s + 1613825)\, y + (-378s + 9102838)$$

Absolute polynomial of L_x over $\mathbf{Q}$ (degree 10):

```
y^10 - 6063*y^8 - 160268*y^7 + 48759770*y^6 + 3120666386*y^5 + 68519
857431*y^4 + 550927965822*y^3 + 4382899360977*y^2 + 20014501791156*y
+ 88050878675188
```

The automorphism σ_x is determined by the image of the primitive element θ: the coordinates of $\sigma_x(\theta)$ in the integral basis of L_x are $(-1 \quad -1 \quad 0 \quad 0 \quad 0 \quad -1 \quad 1 \quad 1 \quad 0 \quad 0)$.

The pair (a', J) representing the torsion class

$$a' = -\frac{302351}{282429536481} + \left(\frac{26}{847288609443}\right) s, \qquad J = \begin{pmatrix} 243 & 146 \\ 0 & 1 \end{pmatrix}.$$

$N(J) = 243$. **Checked:** $(a')\, J^5 = \mathcal{O}_K$, equivalently $\operatorname{div}(a') + 5J = 0$: pass.

The auxiliary pair (t, I')

I' is the ideal of L_x with the following HNF matrix in the integral basis of the absolute model; its absolute norm is $57625467275575 = 5^2 \cdot 895691 \cdot 2573453$:

$$\begin{pmatrix}
11525093455115 & 9445693474315 & 6933200577284 & 6244744795605 & 11442978292550 & 2260349059294 & 9434205862078 & 773797320163 & 629768752977 & 7962598032971 \\
0 & 5 & 0 & 4 & 3 & 0 & 4 & 1 & 0 & 1 \\
0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1
\end{pmatrix}$$

The element t is stored in compact form as a product of 316 factors with signed exponents of absolute value up to 20435587795; it is never expanded (its principal ideal and residues are evaluated factor by factor).

Sign check and certified value

Residue prime: the stored prime ideal above $\ell = 23$ with residue degree 1; there the quotient $N_x(t)/a'$ reduces to the residue of $+1$. Certified value: $D_x(e_1) = (4 \quad 0 \quad 3)$ in the coordinates of $\mathrm{Cl}(K)/5$.

Entry 5: character b, column e_2

Prescribed character vector $(0 \quad 1 \quad 0)$, i.e. the values of x on the classes of the three stored generators of $\mathrm{Cl}(K)$.

The extension and its normalized generator

Relative polynomial of L_x over K:

$$y^5 + (s - 3032)\, y^3 + (36s - 80152)\, y^2 + (341s + 1613825)\, y + (-378s + 9102838)$$

Absolute polynomial of L_x over $\mathbf{Q}$ (degree 10):

```
y^10 - 6063*y^8 - 160268*y^7 + 48759770*y^6 + 3120666386*y^5 + 68519
857431*y^4 + 550927965822*y^3 + 4382899360977*y^2 + 20014501791156*y
+ 88050878675188
```

The automorphism σ_x is determined by the image of the primitive element θ: the coordinates of $\sigma_x(\theta)$ in the integral basis of L_x are $(-1 \quad -1 \quad 0 \quad 0 \quad 0 \quad -1 \quad 1 \quad 1 \quad 0 \quad 0)$.

The pair (a', J) representing the torsion class

$$a' = -\frac{107118472}{297293147712326817} + (\frac{147313}{891879443136980451})\, s, \qquad J = \begin{pmatrix} 3891 & 2759 \\ 0 & 1 \end{pmatrix}.$$

$N(J) = 3891$. **Checked:** $(a')\, J^5 = \mathcal{O}_K$, equivalently $\operatorname{div}(a') + 5J = 0$: pass.

The auxiliary pair (t, I')

I' is the ideal of L_x with the following HNF matrix in the integral basis of the absolute model; its absolute norm is $57625467275575 = 5^2 \cdot 895691 \cdot 2573453$:

$$\begin{pmatrix} 11525093455115 & 9445693474315 & 6933200577284 & 6244744795605 & 11442978292550 & 2260349059294 & 9434205862078 & 773797320163 & 629768752977 & 7962598032971 \\ 0 & 5 & 0 & 4 & 3 & 0 & 4 & 1 & 0 & 1 \\ 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 \end{pmatrix}$$

The element t is stored in compact form as a product of 309 factors with signed exponents of absolute value up to 32788251594; it is never expanded (its principal ideal and residues are evaluated factor by factor).

Sign check and certified value

Residue prime: the stored prime ideal above $\ell = 23$ with residue degree 1; there the quotient $N_x(t)/a'$ reduces to the residue of $+1$. Certified value: $D_x(e_2) = (4 \quad 0 \quad 3)$ in the coordinates of $\mathrm{Cl}(K)/5$.

Entry 6: character b, column e_3

Prescribed character vector $(0 \quad 1 \quad 0)$, i.e. the values of x on the classes of the three stored generators of $\mathrm{Cl}(K)$.

The extension and its normalized generator

Relative polynomial of L_x over K:

$$y^5 + (s - 3032)\, y^3 + (36s - 80152)\, y^2 + (341s + 1613825)\, y + (-378s + 9102838)$$

Absolute polynomial of L_x over $\mathbf{Q}$ (degree 10):

```
y^10 - 6063*y^8 - 160268*y^7 + 48759770*y^6 + 3120666386*y^5 + 68519
857431*y^4 + 550927965822*y^3 + 4382899360977*y^2 + 20014501791156*y
+ 88050878675188
```

The automorphism σ_x is determined by the image of the primitive element θ: the coordinates of $\sigma_x(\theta)$ in the integral basis of L_x are $(-1 \quad -1 \quad 0 \quad 0 \quad 0 \quad -1 \quad 1 \quad 1 \quad 0 \quad 0)$.

The pair (a', J) representing the torsion class

$$a' = -\frac{567178771}{2483696869417001583} + (-\frac{354041}{7451090608251004749})\, s, \qquad J = \begin{pmatrix} 5949 & 3737 \\ 0 & 1 \end{pmatrix}.$$

$N(J) = 5949$. **Checked:** $(a')\, J^5 = \mathcal{O}_K$, equivalently $\operatorname{div}(a') + 5J = 0$: pass.

The auxiliary pair (t, I')

I' is the ideal of L_x with the following HNF matrix in the integral basis of the absolute model; its absolute norm is $139620653811925 = 5^2 \cdot 11^2 \cdot 13 \cdot 1009 \cdot 1297 \cdot 2713$:

$$\begin{pmatrix} 2538557342035 & 2193343278236 & 2412385824361 & 1348386902375 & 2324460976244 & 907558887807 & 1556954595250 & 1019117867398 & 2419996922939 & 2499913022996 \\ 0 & 11 & 6 & 10 & 9 & 5 & 7 & 6 & 2 & 7 \\ 0 & 0 & 5 & 3 & 4 & 2 & 4 & 0 & 0 & 4 \\ 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 \end{pmatrix}$$

The element t is stored in compact form as a product of 314 factors with signed exponents of absolute value up to 32997314158; it is never expanded (its principal ideal and residues are evaluated factor by factor).

Sign check and certified value

Residue prime: the stored prime ideal above $\ell = 23$ with residue degree 1; there the quotient $N_x(t)/a'$ reduces to the residue of $+1$. Certified value: $D_x(e_3) = (3 \quad 0 \quad 0)$ in the coordinates of $\mathrm{Cl}(K)/5$.

Entry 7: character c, column e_1

Prescribed character vector $(0 \quad 0 \quad 1)$, i.e. the values of x on the classes of the three stored generators of $\mathrm{Cl}(K)$.

The extension and its normalized generator

Relative polynomial of L_x over K:

$$y^5 + 6193y^3 + (-12s + 6)\, y^2 + 11953708y + (-17232s + 8616)$$

Absolute polynomial of L_x over $\mathbf{Q}$ (degree 10):

```
y^10 + 12386*y^8 + 62260665*y^6 + 153291844580*y^4 + 157920935011888
*y^2 + 10791396444964032
```

The automorphism σ_x is determined by the image of the primitive element θ: the coordinates of $\sigma_x(\theta)$ in the integral basis of L_x are $(0 \quad 0 \quad 1 \quad -1 \quad 1 \quad 0 \quad 0 \quad 0 \quad 0 \quad 0)$.

The pair (a', J) representing the torsion class

$$a' = -\frac{302351}{282429536481} + \left(\frac{26}{847288609443}\right) s, \qquad J = \begin{pmatrix} 243 & 146 \\ 0 & 1 \end{pmatrix}.$$

$N(J) = 243$. **Checked:** $(a')\, J^5 = \mathcal{O}_K$, equivalently $\mathrm{div}(a') + 5J = 0$: pass.

The auxiliary pair (t, I')

I' is the ideal of L_x with the following HNF matrix in the integral basis of the absolute model; its absolute norm is $733382257797396 = 2^2 \cdot 3^4 \cdot 29^2 \cdot 33829 \cdot 79561$:

$$\begin{pmatrix}
1404946854018 & 154977889524 & 940265332011 & 956854072224 & 719905291077 & 158133111898 & 865475666966 & 131303827082 & 1345682429978 & 398580538653 \\
0 & 87 & 3 & 36 & 6 & 35 & 15 & 83 & 58 & 48 \\
0 & 0 & 3 & 2 & 1 & 2 & 0 & 0 & 0 & 1 \\
0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 2 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1
\end{pmatrix}$$

The element t is stored in compact form as a product of 229 factors with signed exponents of absolute value up to 3626337429; it is never expanded (its principal ideal and residues are evaluated factor by factor).

Sign check and certified value

Residue prime: the stored prime ideal above $\ell = 13$ with residue degree 1; there the quotient $N_x(t)/a'$ reduces to the residue of $+1$. Certified value: $D_x(e_1) = (3 \quad 0 \quad 0)$ in the coordinates of $\mathrm{Cl}(K)/5$.

Entry 8: character c, column e_2

Prescribed character vector $(0 \quad 0 \quad 1)$, i.e. the values of x on the classes of the three stored generators of $\mathrm{Cl}(K)$.

The extension and its normalized generator

Relative polynomial of L_x over K:

$$y^5 + 6193y^3 + (-12s + 6)\,y^2 + 11953708y + (-17232s + 8616)$$

Absolute polynomial of L_x over $\mathbf{Q}$ (degree 10):

```
y^10 + 12386*y^8 + 62260665*y^6 + 153291844580*y^4 + 157920935011888
*y^2 + 10791396444964032
```

The automorphism σ_x is determined by the image of the primitive element θ: the coordinates of $\sigma_x(\theta)$ in the integral basis of L_x are $(0 \quad 0 \quad 1 \quad -1 \quad 1 \quad 0 \quad 0 \quad 0 \quad 0 \quad 0)$.

The pair (a', J) representing the torsion class

$$a' = -\frac{107118472}{297293147712326817} + \left(\frac{147313}{891879443136980451}\right) s, \qquad J = \begin{pmatrix} 3891 & 2759 \\ 0 & 1 \end{pmatrix}.$$

$N(J) = 3891$. **Checked:** $(a')\,J^5 = \mathcal{O}_K$, equivalently $\mathrm{div}(a') + 5J = 0$: pass.

The auxiliary pair (t, I')

I' is the ideal of L_x with the following HNF matrix in the integral basis of the absolute model; its absolute norm is $18686585256372 = 2^2 \cdot 3^5 \cdot 11 \cdot 1747716541$:

$$\begin{pmatrix}
346047875118 & 334467155754 & 84740082138 & 72503129515 & 207783862989 & 204701488444 & 307652418027 & 215616585110 & 134219496680 & 175077894901 \\
0 & 9 & 3 & 0 & 0 & 1 & 8 & 5 & 2 & 3 \\
0 & 0 & 3 & 1 & 1 & 2 & 2 & 0 & 2 & 0 \\
0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 2 & 1 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1
\end{pmatrix}$$

The element t is stored in compact form as a product of 224 factors with signed exponents of absolute value up to 2074955016; it is never expanded (its principal ideal and residues are evaluated factor by factor).

Sign check and certified value

Residue prime: the stored prime ideal above $\ell = 13$ with residue degree 1; there the quotient $N_x(t)/a'$ reduces to the residue of $+1$. Certified value: $D_x(e_2) = (3 \quad 0 \quad 0)$ in the coordinates of $\mathrm{Cl}(K)/5$.

Entry 9: character c, column e_3

Prescribed character vector $(0 \quad 0 \quad 1)$, i.e. the values of x on the classes of the three stored generators of $\mathrm{Cl}(K)$.

The extension and its normalized generator

Relative polynomial of L_x over K:

$$y^5 + 6193y^3 + (-12s + 6)\, y^2 + 11953708y + (-17232s + 8616)$$

Absolute polynomial of L_x over $\mathbf{Q}$ (degree 10):

```
y^10 + 12386*y^8 + 62260665*y^6 + 153291844580*y^4 + 157920935011888
*y^2 + 10791396444964032
```

The automorphism σ_x is determined by the image of the primitive element θ: the coordinates of $\sigma_x(\theta)$ in the integral basis of L_x are $(0 \quad 0 \quad 1 \quad -1 \quad 1 \quad 0 \quad 0 \quad 0 \quad 0 \quad 0)$.

The pair (a', J) representing the torsion class

$$a' = -\frac{567178771}{2483696869417001583} + \left(-\frac{354041}{7451090608251004749}\right) s, \qquad J = \begin{pmatrix} 5949 & 3737 \\ 0 & 1 \end{pmatrix}.$$

$N(J) = 5949$. **Checked:** $(a')\, J^5 = \mathcal{O}_K$, equivalently $\operatorname{div}(a') + 5J = 0$: pass.

The auxiliary pair (t, I')

I' is the ideal of L_x with the following HNF matrix in the integral basis of the absolute model; its absolute norm is $33273839753748 = 2^2 \cdot 3^4 \cdot 11^2 \cdot 29 \cdot 29266753$:

$$\begin{pmatrix}
168049695726 & 107035048428 & 139837805217 & 154742764422 & 156952346499 & 127293816998 & 129365504829 & 140880265544 & 55121356777 & 102705190209 \\
0 & 33 & 21 & 30 & 30 & 3 & 29 & 31 & 7 & 30 \\
0 & 0 & 3 & 2 & 1 & 2 & 1 & 2 & 1 & 1 \\
0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 2 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1
\end{pmatrix}$$

The element t is stored in compact form as a product of 225 factors with signed exponents of absolute value up to 9842927219; it is never expanded (its principal ideal and residues are evaluated factor by factor).

Sign check and certified value

Residue prime: the stored prime ideal above $\ell = 13$ with residue degree 1; there the quotient $N_x(t)/a'$ reduces to the residue of $+1$. Certified value: $D_x(e_3) = (3 \quad 0 \quad 0)$ in the coordinates of $\mathrm{Cl}(K)/5$.

Entry 10: character $a + b$, column e_1

Prescribed character vector $(1 \quad 1 \quad 0)$, i.e. the values of x on the classes of the three stored generators of $\mathrm{Cl}(K)$.

The extension and its normalized generator

Relative polynomial of L_x over K:

$$y^5 - 2y^4 + (s - 47)\, y^3 + (-32s - 65038)\, y^2 + (569s + 347968)\, y + (-6358s - 6782218)$$

Absolute polynomial of L_x over $\mathbf{Q}$ (degree 10):

```
y^10 - 4*y^9 - 89*y^8 - 129922*y^7 + 37300670*y^6 - 2334788134*y^5 +
82797719975*y^4 - 1830224060950*y^3 + 27558076586615*y^2 - 26767357
2155582*y + 1515127960082636
```

The automorphism σ_x is determined by the image of the primitive element θ: the coordinates of $\sigma_x(\theta)$ in the integral basis of L_x are $(0 \quad 0 \quad 0 \quad 1 \quad 0 \quad 0 \quad 1 \quad 0 \quad 0 \quad 0)$.

The pair (a', J) representing the torsion class

$$a' = -\frac{302351}{282429536481} + \Big(\frac{26}{847288609443}\Big)\, s, \qquad J = \begin{pmatrix} 243 & 146 \\ 0 & 1 \end{pmatrix}.$$

$N(J) = 243$. **Checked:** $(a')\, J^5 = \mathcal{O}_K$, equivalently $\mathrm{div}(a') + 5J = 0$: pass.

The auxiliary pair (t, I')

I' is the ideal of L_x with the following HNF matrix in the integral basis of the absolute model; its absolute norm is $1125460241408 = 2^{10} \cdot 59 \cdot 18628513$:

$$\begin{pmatrix}
17585316272 & 419466988 & 5022817258 & 6603836502 & 15893624172 & 5209419996 & 12971527225 & 5013576814 & 6548689990 & 4555065871 \\
0 & 4 & 0 & 2 & 1 & 2 & 3 & 2 & 1 & 0 \\
0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 4 & 2 & 3 & 1 & 2 & 2 & 2 \\
0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 2 & 0 & 0 & 1 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 2 & 0 & 1 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1
\end{pmatrix}$$

The element t is stored in compact form as a product of 7 factors with signed exponents of absolute value up to 2; it is never expanded (its principal ideal and residues are evaluated factor by factor).

Sign check and certified value

Residue prime: the stored prime ideal above $\ell = 3$ with residue degree 1; there the quotient $N_x(t)/a'$ reduces to the residue of $+1$. Certified value: $D_x(e_1) = (2 \quad 3 \quad 1)$ in the coordinates of $\mathrm{Cl}(K)/5$.

Entry 11: character $a + b$, column e_2

Prescribed character vector $(1 \quad 1 \quad 0)$, i.e. the values of x on the classes of the three stored generators of $\mathrm{Cl}(K)$.

The extension and its normalized generator

Relative polynomial of L_x over K:

$$y^5 - 2y^4 + (s - 47)\, y^3 + (-32s - 65038)\, y^2 + (569s + 347968)\, y + (-6358s - 6782218)$$

Absolute polynomial of L_x over $\mathbf{Q}$ (degree 10):

```
y^10 - 4*y^9 - 89*y^8 - 129922*y^7 + 37300670*y^6 - 2334788134*y^5 +
82797719975*y^4 - 1830224060950*y^3 + 27558076586615*y^2 - 26767357
2155582*y + 1515127960082636
```

The automorphism σ_x is determined by the image of the primitive element θ: the coordinates of $\sigma_x(\theta)$ in the integral basis of L_x are $(0 \quad 0 \quad 0 \quad 1 \quad 0 \quad 0 \quad 1 \quad 0 \quad 0 \quad 0)$.

The pair (a', J) representing the torsion class

$$a' = -\frac{107118472}{297293147712326817} + \left(\frac{147313}{891879443136980451}\right) s, \qquad J = \begin{pmatrix} 3891 & 2759 \\ 0 & 1 \end{pmatrix}.$$

$N(J) = 3891$. **Checked:** $(a')\, J^5 = \mathcal{O}_K$, equivalently $\operatorname{div}(a') + 5J = 0$: pass.

The auxiliary pair (t, I')

I' is the ideal of L_x with the following HNF matrix in the integral basis of the absolute model; its absolute norm is $1553187443925056 = 2^6 \cdot 31^2 \cdot 59 \cdot 428024371$:

$$\begin{pmatrix}
6262852596472 & 4952689442050 & 3385043123645 & 5262592009628 & 4682207914831 & 2454240298106 & 3080031937494 & 2478168062243 & 4658895014553 & 3445007334746 \\
0 & 62 & 31 & 26 & 31 & 22 & 32 & 42 & 31 & 27 \\
0 & 0 & 2 & 0 & 0 & 1 & 1 & 0 & 1 & 1 \\
0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 2 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1
\end{pmatrix}$$

The element t is stored in compact form as a product of 7 factors with signed exponents of absolute value up to 2; it is never expanded (its principal ideal and residues are evaluated factor by factor).

Sign check and certified value

Residue prime: the stored prime ideal above $\ell = 3$ with residue degree 1; there the quotient $N_x(t)/a'$ reduces to the residue of $+1$. Certified value: $D_x(e_2) = (2 \quad 3 \quad 4)$ in the coordinates of $\mathrm{Cl}(K)/5$.

Entry 12: character $a + b$, column e_3

Prescribed character vector $(1 \quad 1 \quad 0)$, i.e. the values of x on the classes of the three stored generators of $\mathrm{Cl}(K)$.

The extension and its normalized generator

Relative polynomial of L_x over K:

$$y^5 - 2y^4 + (s - 47)\, y^3 + (-32s - 65038)\, y^2 + (569s + 347968)\, y + (-6358s - 6782218)$$

Absolute polynomial of L_x over $\mathbf{Q}$ (degree 10):

```
y^10 - 4*y^9 - 89*y^8 - 129922*y^7 + 37300670*y^6 - 2334788134*y^5 +
82797719975*y^4 - 1830224060950*y^3 + 27558076586615*y^2 - 26767357
2155582*y + 1515127960082636
```

The automorphism σ_x is determined by the image of the primitive element θ: the coordinates of $\sigma_x(\theta)$ in the integral basis of L_x are $(0 \quad 0 \quad 0 \quad 1 \quad 0 \quad 0 \quad 1 \quad 0 \quad 0 \quad 0)$.

The pair (a', J) representing the torsion class

$$a' = -\frac{567178771}{2483696869417001583} + \left(-\frac{354041}{7451090608251004749}\right) s, \qquad J = \begin{pmatrix} 5949 & 3737 \\ 0 & 1 \end{pmatrix}.$$

$N(J) = 5949$. **Checked:** $(a')\, J^5 = \mathcal{O}_K$, equivalently $\operatorname{div}(a') + 5J = 0$: pass.

The auxiliary pair (t, I')

I' is the ideal of L_x with the following HNF matrix in the integral basis of the absolute model; its absolute norm is $1114015603054132 = 2^2 \cdot 59 \cdot 16381 \cdot 288163427$:

$$\begin{pmatrix} 557007801527066 & 316934237403910 & 187216643674956 & 285774196366499 & 258780131924718 & 38327651433291 & 473207594079892 & 511997291929061 & 438155034157406 & 384728200116386 \\ 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 2 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 \end{pmatrix}$$

The element t is stored in compact form as a product of 7 factors with signed exponents of absolute value up to 2; it is never expanded (its principal ideal and residues are evaluated factor by factor).

Sign check and certified value

Residue prime: the stored prime ideal above $\ell = 3$ with residue degree 1; there the quotient $N_x(t)/a'$ reduces to the residue of $+1$. Certified value: $D_x(e_3) = (3 \quad 2 \quad 4)$ in the coordinates of $\mathrm{Cl}(K)/5$.

Entry 13: character $a + c$, column e_1

Prescribed character vector $(1 \quad 0 \quad 1)$, i.e. the values of x on the classes of the three stored generators of $\mathrm{Cl}(K)$.

The extension and its normalized generator

Relative polynomial of L_x over K:

$$y^5 - 2y^4 + 3908y^3 + (-18s - 256495)\, y^2 + (1278s - 2337894)\, y + (-18144s - 84030696)$$

Absolute polynomial of L_x over $\mathbf{Q}$ (degree 10):

```
y^10 - 4*y^9 + 7820*y^8 - 528640*y^7 + 11623970*y^6 - 2163565780*y^5
+ 59637214915*y^4 - 1129837238442*y^3 + 131670166703868*y^2 - 12925
44292278672*y + 19026570918295872
```

The automorphism σ_x is determined by the image of the primitive element θ: the coordinates of $\sigma_x(\theta)$ in the integral basis of L_x are $(1 \quad 0 \quad 0 \quad -1 \quad 0 \quad 1 \quad 0 \quad 0 \quad -1 \quad 0)$.

The pair (a', J) representing the torsion class

$$a' = -\frac{302351}{282429536481} + (\frac{26}{847288609443})\, s, \qquad J = \begin{pmatrix} 243 & 146 \\ 0 & 1 \end{pmatrix}.$$

$N(J) = 243$. **Checked:** $(a')\, J^5 = \mathcal{O}_K$, equivalently $\operatorname{div}(a') + 5J = 0$: pass.

The auxiliary pair (t, I')

I' is the ideal of L_x with the following HNF matrix in the integral basis of the absolute model; its absolute norm is $831195214191 = 3^6 \cdot 1140185479$:

$$\begin{pmatrix}
10261669311 & 1981050183 & 4044114609 & 2075129937 & 141688725 & 853100163 & 9381716019 & 2390399262 & 8743021250 & 9683745395 \\
0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 3 & 1 & 0 & 0 & 0 & 0 & 1 & 2 \\
0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 3 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 3 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 3 & 0 & 1 & 1 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1
\end{pmatrix}$$

The element t is stored in compact form as a product of 354 factors with signed exponents of absolute value up to 368331516325253843; it is never expanded (its principal ideal and residues are evaluated factor by factor).

Sign check and certified value

Residue prime: the stored prime ideal above $\ell = 11$ with residue degree 1; there the quotient $N_x(t)/a'$ reduces to the residue of $+1$. Certified value: $D_x(e_1) = (2 \quad 4 \quad 3)$ in the coordinates of $\mathrm{Cl}(K)/5$.

Entry 14: character $a + c$, column e_2

Prescribed character vector $(1 \quad 0 \quad 1)$, i.e. the values of x on the classes of the three stored generators of $\mathrm{Cl}(K)$.

The extension and its normalized generator

Relative polynomial of L_x over K:

$$y^5 - 2y^4 + 3908y^3 + (-18s - 256495)\, y^2 + (1278s - 2337894)\, y + (-18144s - 84030696)$$

Absolute polynomial of L_x over $\mathbf{Q}$ (degree 10):

```
y^10 - 4*y^9 + 7820*y^8 - 528640*y^7 + 11623970*y^6 - 2163565780*y^5
+ 59637214915*y^4 - 1129837238442*y^3 + 131670166703868*y^2 - 12925
44292278672*y + 19026570918295872
```

The automorphism σ_x is determined by the image of the primitive element θ: the coordinates of $\sigma_x(\theta)$ in the integral basis of L_x are $(1 \quad 0 \quad 0 \quad -1 \quad 0 \quad 1 \quad 0 \quad 0 \quad -1 \quad 0)$.

The pair (a', J) representing the torsion class

$$a' = -\frac{107118472}{297293147712326817} + (\frac{147313}{891879443136980451})\, s, \qquad J = \begin{pmatrix} 3891 & 2759 \\ 0 & 1 \end{pmatrix}.$$

$N(J) = 3891$. **Checked:** $(a')\, J^5 = \mathcal{O}_K$, equivalently $\operatorname{div}(a') + 5J = 0$: pass.

The auxiliary pair (t, I')

I' is the ideal of L_x with the following HNF matrix in the integral basis of the absolute model; its absolute norm is $15351132355983 = 3^7 \cdot 103 \cdot 1303 \cdot 52301$:

$$\begin{pmatrix} 189520152543 & 20575260729 & 58767465024 & 145265466492 & 165369457578 & 111040597788 & 114758162932 & 37286570917 & 67241036120 & 134200016196 \\ 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 9 & 3 & 0 & 1 & 2 & 3 & 3 \\ 0 & 0 & 0 & 0 & 3 & 0 & 2 & 2 & 1 & 2 \\ 0 & 0 & 0 & 0 & 0 & 3 & 2 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 \end{pmatrix}$$

The element t is stored in compact form as a product of 360 factors with signed exponents of absolute value up to 493389506105579070; it is never expanded (its principal ideal and residues are evaluated factor by factor).

Sign check and certified value

Residue prime: the stored prime ideal above $\ell = 11$ with residue degree 1; there the quotient $N_x(t)/a'$ reduces to the residue of $+1$. Certified value: $D_x(e_2) = (0 \quad 1 \quad 0)$ in the coordinates of $\mathrm{Cl}(K)/5$.

Entry 15: character $a + c$, column e_3

Prescribed character vector $(1 \quad 0 \quad 1)$, i.e. the values of x on the classes of the three stored generators of $\mathrm{Cl}(K)$.

The extension and its normalized generator

Relative polynomial of L_x over K:

$$y^5 - 2y^4 + 3908y^3 + (-18s - 256495)\, y^2 + (1278s - 2337894)\, y + (-18144s - 84030696)$$

Absolute polynomial of L_x over $\mathbf{Q}$ (degree 10):

```
y^10 - 4*y^9 + 7820*y^8 - 528640*y^7 + 11623970*y^6 - 2163565780*y^5
+ 59637214915*y^4 - 1129837238442*y^3 + 131670166703868*y^2 - 12925
44292278672*y + 19026570918295872
```

The automorphism σ_x is determined by the image of the primitive element θ: the coordinates of $\sigma_x(\theta)$ in the integral basis of L_x are $(1 \quad 0 \quad 0 \quad -1 \quad 0 \quad 1 \quad 0 \quad 0 \quad -1 \quad 0)$.

The pair (a', J) representing the torsion class

$$a' = -\frac{567178771}{2483696869417001583} + \left(-\frac{354041}{7451090608251004749}\right) s, \qquad J = \begin{pmatrix} 5949 & 3737 \\ 0 & 1 \end{pmatrix}.$$

$N(J) = 5949$. **Checked:** $(a')\, J^5 = \mathcal{O}_K$, equivalently $\mathrm{div}(a') + 5J = 0$: pass.

The auxiliary pair (t, I')

I' is the ideal of L_x with the following HNF matrix in the integral basis of the absolute model; its absolute norm is $859350175010928 = 2^4 \cdot 3^{10} \cdot 367 \cdot 1319 \cdot 1879$:

$$
\begin{pmatrix}
49116951018 & 16467902424 & 14811877458 & 43459052580 & 37915200955 & 7014715698 & 24982371281 & 10672102490 & 47978491072 & 47515955827 \\
0 & 18 & 0 & 9 & 0 & 6 & 6 & 14 & 2 & 15 \\
0 & 0 & 6 & 3 & 5 & 0 & 2 & 2 & 4 & 4 \\
0 & 0 & 0 & 27 & 1 & 6 & 2 & 6 & 2 & 25 \\
0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 6 & 2 & 4 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1
\end{pmatrix}
$$

The element t is stored in compact form as a product of 355 factors with signed exponents of absolute value up to 71692846783669673; it is never expanded (its principal ideal and residues are evaluated factor by factor).

Sign check and certified value

Residue prime: the stored prime ideal above $\ell = 11$ with residue degree 1; there the quotient $N_x(t)/a'$ reduces to the residue of $+1$. Certified value: $D_x(e_3) = (3 \quad 0 \quad 2)$ in the coordinates of $\mathrm{Cl}(K)/5$.

Entry 16: character $b + c$, column e_1

Prescribed character vector $(0 \quad 1 \quad 1)$, i.e. the values of x on the classes of the three stored generators of $\mathrm{Cl}(K)$.

The extension and its normalized generator

Relative polynomial of L_x over K:

$$y^5 - y^4 + (-s - 5451)\, y^3 + (55s + 197846)\, y^2 + (-751s - 3761924)\, y + (1488s + 51286436)$$

Absolute polynomial of L_x over $\mathbf{Q}$ (degree 10):

```
y^10 - 2*y^9 - 10902*y^8 + 406650*y^7 + 58140293*y^6 - 6044912354*y^
5 + 244590967708*y^4 - 5158451024552*y^3 + 60896872875187*y^2 - 4671
38484524188*y + 2710840781434192
```

The automorphism σ_x is determined by the image of the primitive element θ: the coordinates of $\sigma_x(\theta)$ in the integral basis of L_x are $(2 \quad 1 \quad -1 \quad 1 \quad 1 \quad 0 \quad 0 \quad 1 \quad 1 \quad 0)$.

The pair (a', J) representing the torsion class

$$a' = -\frac{302351}{282429536481} + (\frac{26}{847288609443})\, s, \qquad J = \begin{pmatrix} 243 & 146 \\ 0 & 1 \end{pmatrix}.$$

$N(J) = 243$. **Checked:** $(a')\, J^5 = \mathcal{O}_K$, equivalently $\mathrm{div}(a') + 5J = 0$: pass.

The auxiliary pair (t, I')

I' is the ideal of L_x with the following HNF matrix in the integral basis of the absolute model; its absolute norm is $1077847310102068 = 2^2 \cdot 11^2 \cdot 1334563 \cdot 1668679$:

$$
\begin{pmatrix}
538923655051034 & 278237979419006 & 190485077547198 & 421419002106049 & 84552641985817 & 351988813244474 & 306920509858934 & 521180535255022 & 12921973035717 & 162488895613597 \\
0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 2 & 1 & 0 & 1 & 1 & 0 & 1 & 1 \\
0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1
\end{pmatrix}
$$

The element t is stored in compact form as a product of 475 factors with signed exponents of absolute value up to 96091855382639210730; it is never expanded (its principal ideal and residues are evaluated factor by factor).

Sign check and certified value

Residue prime: the stored prime ideal above $\ell = 13$ with residue degree 1; there the quotient $N_x(t)/a'$ reduces to the residue of $+1$. Certified value: $D_x(e_1) = (3 \quad 2 \quad 3)$ in the coordinates of $\mathrm{Cl}(K)/5$.

Entry 17: character $b + c$, column e_2

Prescribed character vector $(0 \quad 1 \quad 1)$, i.e. the values of x on the classes of the three stored generators of $\mathrm{Cl}(K)$.

The extension and its normalized generator

Relative polynomial of L_x over K:

$$y^5 - y^4 + (-s - 5451)\, y^3 + (55s + 197846)\, y^2 + (-751s - 3761924)\, y + (1488s + 51286436)$$

Absolute polynomial of L_x over $\mathbf{Q}$ (degree 10):

```
y^10 - 2*y^9 - 10902*y^8 + 406650*y^7 + 58140293*y^6 - 6044912354*y^
5 + 244590967708*y^4 - 5158451024552*y^3 + 60896872875187*y^2 - 4671
38484524188*y + 2710840781434192
```

The automorphism σ_x is determined by the image of the primitive element θ: the coordinates of $\sigma_x(\theta)$ in the integral basis of L_x are $(2 \quad 1 \quad -1 \quad 1 \quad 1 \quad 0 \quad 0 \quad 1 \quad 1 \quad 0)$.

The pair (a', J) representing the torsion class

$$a' = -\frac{107118472}{297293147712326817} + \left(\frac{147313}{891879443136980451}\right) s, \qquad J = \begin{pmatrix} 3891 & 2759 \\ 0 & 1 \end{pmatrix}.$$

$N(J) = 3891$. **Checked:** $(a')\, J^5 = \mathcal{O}_K$, equivalently $\mathrm{div}(a') + 5J = 0$: pass.

The auxiliary pair (t, I')

I' is the ideal of L_x with the following HNF matrix in the integral basis of the absolute model; its absolute norm is $704056868913653 = 1249511 \cdot 563465923$:

$$\begin{pmatrix} 704056868913653 & 677929875813929 & 86723393268919 & 563709172198755 & 322507934507511 & 630525880570571 & 586832490503761 & 661861694017147 & 245146328672023 & 386453179794760 \\ 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 \end{pmatrix}$$

The element t is stored in compact form as a product of 480 factors with signed exponents of absolute value up to 114222208111073801679; it is never expanded (its principal ideal and residues are evaluated factor by factor).

Sign check and certified value

Residue prime: the stored prime ideal above $\ell = 13$ with residue degree 1; there the quotient $N_x(t)/a'$ reduces to the residue of $+1$. Certified value: $D_x(e_2) = (0 \quad 2 \quad 3)$ in the coordinates of $\mathrm{Cl}(K)/5$.

Entry 18: character $b + c$, column e_3

Prescribed character vector $(0 \quad 1 \quad 1)$, i.e. the values of x on the classes of the three stored generators of $\mathrm{Cl}(K)$.

The extension and its normalized generator

Relative polynomial of L_x over K:

$$y^5 - y^4 + (-s - 5451)\, y^3 + (55s + 197846)\, y^2 + (-751s - 3761924)\, y + (1488s + 51286436)$$

Absolute polynomial of L_x over $\mathbf{Q}$ (degree 10):

```
y^10 - 2*y^9 - 10902*y^8 + 406650*y^7 + 58140293*y^6 - 6044912354*y^
5 + 244590967708*y^4 - 5158451024552*y^3 + 60896872875187*y^2 - 4671
38484524188*y + 2710840781434192
```

The automorphism σ_x is determined by the image of the primitive element θ: the coordinates of $\sigma_x(\theta)$ in the integral basis of L_x are $(2 \quad 1 \quad -1 \quad 1 \quad 1 \quad 0 \quad 0 \quad 1 \quad 1 \quad 0)$.

The pair (a', J) representing the torsion class

$$a' = -\frac{567178771}{2483696869417001583} + (-\frac{354041}{7451090608251004749})\, s, \qquad J = \begin{pmatrix} 5949 & 3737 \\ 0 & 1 \end{pmatrix}.$$

$N(J) = 5949$. **Checked:** $(a')\, J^5 = \mathcal{O}_K$, equivalently $\mathrm{div}(a') + 5J = 0$: pass.

The auxiliary pair (t, I')

I' is the ideal of L_x with the following HNF matrix in the integral basis of the absolute model; its absolute norm is $357105834361012 = 2^2 \cdot 11 \cdot 79^3 \cdot 16461257$:

$$\begin{pmatrix}
28609664666 & 25753292476 & 19995026400 & 4706669011 & 4624920681 & 585410048 & 12633787598 & 18175140166 & 9766273491 & 25609712223 \\
0 & 79 & 0 & 1 & 49 & 65 & 29 & 14 & 0 & 21 \\
0 & 0 & 158 & 127 & 78 & 69 & 89 & 98 & 139 & 35 \\
0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1
\end{pmatrix}$$

The element t is stored in compact form as a product of 476 factors with signed exponents of absolute value up to 53839552848120232929; it is never expanded (its principal ideal and residues are evaluated factor by factor).

Sign check and certified value

Residue prime: the stored prime ideal above $\ell = 13$ with residue degree 1; there the quotient $N_x(t)/a'$ reduces to the residue of $+1$. Certified value: $D_x(e_3) = (2 \quad 0 \quad 0)$ in the coordinates of $\mathrm{Cl}(K)/5$.